Datatypes

“JavaScript is a dynamic type language, means you don't need to specify type of the variable because it is dynamically used by JavaScript engine. ”

Datatypes

- JavaScript is **dynamically typed** (variables can change types).
- Two main categories:
  - **Primitive** (immutable, stored by value)
  - **Non-Primitive** (mutable, stored by reference)

JavaScript Data Types

Primitive Types (by Value)

- String
- Number
- Boolean
- Null
- Undefined
- Symbol
- BigInt

Non-Primitive Types (by Reference)

- Object → {}
- Array → []
- Function → function() {}
- Date, RegExp, Map, Set, etc.

Primitive data type

- The predefined data types provided by JavaScript language are known as primitive data types. Primitive data types are also known as **in-built** data types.

Data Type	Description	Example
Number	Represents both integer and float values	let age = 25;
String	Represents a sequence of characters	let name = “Mili";
Boolean	Logical value: true or false	let isLoggedIn = true;
Undefined	A variable declared but not assigned	let x;
Null	Represents an intentional absence of value	let data = null;
BigInt	For integers larger than 2^53 - 1	let big = 123n;
Symbol	Unique and immutable identifiers	let sym = Symbol();

Non Primitive data type

- The data types that are derived from primitive data types of the JavaScript language are known as non-primitive data types. It is also known as **derived data types** or **reference** data types.

Data Type	Description	Example
Object	Collection of key-value pairs	let user = { name: “Milio" };
Array	Ordered list of values	let fruits = ["apple", "mango"];
Function	Block of code designed to perform a task	function greet() { }

1 String

- String is used to store text.
- In JavaScript, strings are surrounded by quotes: **Single quotes** and **double quotes** are practically the same and you can use either of them.
- **Backticks** are generally used when you need to include variables or expressions into a string.

Single quotes: 'Hello'

Double quotes: "Hello"

Backticks: `Hello`

```
var name1 = "ram";

var name2 = 'hari';

console.log( `The names are ${name1} and ${name2}`);
```

String Concatenation

1. **String Concatenation using +**

```
let companyName = 'Luminar Technolab'
let place = 'Kochi‘
console.log("My company name is " + company + " and location is " + location);
```

2. **Comma-separated console output**

```
console.log("My company name is ", company, " and location is ", location);
```

3. **String Interpolation using Template Literals (**Modern & Recommended**)**

```
console.log(`My company name is ${company} and location is ${location}`);
```

2 Number

- Number represents integer and floating numbers (decimals and exponentials).

For example,

```
var n1 = 3;

var n2 = 3.433;

var n3 = 3e5                    // 3 * 10^5

console.log(n1, n2, n3);
```

3 Boolean

- This data type represents logical entities.
- Boolean represents one of two values: true or false.
- It is easier to think of it as a yes/no switch.

For example,

```
const dataChecked = true;

const valueCounted = false;
```

4 undefined

- The undefined data type represents value that is not assigned.
- If a variable is declared but the value is not assigned, then the value of that variable will be undefined.
- For example,

```
let name1;

console.log(name1); // undefined
```

5 null

- In JavaScript, null is a special value that represents empty or unknown value.
- For example,
- const number = null

6 BigInt type

- BigInt is a built-in object in JavaScript that provides a way to represent whole numbers larger than 253-1.
- The largest number that JavaScript can reliably represent with the Number primitive is 253-1, which is represented by the MAX_SAFE_INTEGER constant.

```
let bigBin = BigInt("0b10101010010101010011111111111111111");
// 11430854655n
console.log(bigBin);
```

7 Symbols

- Symbols are new primitive built-in object types introduced as part of ES6.
- Symbols return unique identifiers that can be used to add unique property keys to an object that won’t collide with keys of any other code that might add to the object.
- They are used as object properties that cannot be recreated.
- It basically helps us to enable encapsulation or information hiding.

Symbol

```
let symbol1 = Symbol(“Welcome")
let symbol2 = Symbol(“Welcome")

// Each time Symbol() method
// is used to create new global Symbol
console.log(symbol1 == symbol2); // False
```

Non-Primitive Types

- **Object** (basic object - Collection of key-value pairs)
- **Array** (special type of object - List of values)
- **Function** (callable object - Reusable block of code (also an object))
- **Date** , **RegExp**, **Map, Set, WeakMap, WeakSet** (Built-in objects)

Object

- An object is a complex data type that allows us to store collections of data.

For example,

```
const student = {
        firstName: 'ram',
        lastName: null,
         class: 10
};
console.log(student);
```

Array

- An array can hold many values under a single name, and you can access the values by referring to an index number.
- Syntax:

```
const array_name = [item1, item2, ...];
```

Eg:

```
const cars = ["Saab", "Volvo", "BMW“,34354,656546,true,null];
console.log(cars)
```

Function

- Its a block of code designed to perform a particular task.
- A function is executed when "something" invokes it

Syntax

```
function name(parameter1, parameter2, parameter3) {
        // code to be executed
        }
```

Eg

```
function sum(a, b){
     var c = a+b;
     return c;
      }
 console.log(sum(2,2));
```

Explicit vs Implicit Type Conversion

What is Type Conversion?

- Changing a value from one data type to another
- Two approaches:
  - **Implicit (Coercion)**: Automatic by JavaScript
  - **Explicit**: Manual conversion by developer

```
let num = "5";  // string
let result = num * 2; // implicit → 10 (number)
let explicitNum = Number(num); // explicit → 5 (number)
```

TASKS

1) var Name="Max” , var age=34;

//display console as "Hi...My name is Max...34 years old..."

2) Write a program to swap two numbers together

var n1=6

var n2=7

before swap n1=6 n2=7

After swap n1=7 n2=6

TASK

Instructions:

- For **each of the following values**, apply the following JavaScript functions convert the values:
- Number()
- parseInt()
- parseFloat()
- String()
- Boolean()
- isNaN(Number())
- Use console.log() to print each result**.**

TASKS

1. What is JavaScript's primary role in web development?
2. Name some non-browser environments that use JavaScript?
3. Describe JavaScript as a prototype-based language.?
4. In what sense is JavaScript considered a multi-paradigm language?
5. Explain the term "dynamic language" in the context of JavaScript.?
6. Differentiate between imperative and declarative programming styles in JavaScript.?
7. Explain Date, RegExp, Map, and Set in JavaScript with examples.
8. Create a new date object showing the current date and time.
9. Replace all vowels in a string with "*". Using RegExp.
   Input: "Javascript" → Output: "J*v*scr*pt“
10. Create a Map with three key-value pairs: "name", "age", and "course".
11. Create a Set and add values: 10, 20, 30, 10, 20.
    Print the set.
12. Write a function that removes duplicates from an array using Set.
13. Convert the string "123" to a number using three different methods using type conversion.
14. Why does "5" + 2 return "52" while "5" - 2 returns 3? Write code to fix the addition to output 7.